

A Report on:

Improper Disposal of Electronic Waste

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Abstract

Electronic waste (e-waste) is one of the fastest-growing waste streams worldwide. Rapid technological development, frequent replacement of electronic products and increasing consumption generate large

quantities of discarded phones, computers, batteries, chargers, appliances and other electronic equipment. The Global E-waste Monitor 2024 reported that 62 million tonnes of e-waste were generated globally in 2022, while only 22.3% was documented as formally collected and recycled in an environmentally sound manner. The same report projects approximately 82 million tonnes by 2030 if current trends continue [1].

A major challenge is that e-waste collection is often reactive. Collection agencies may not have timely information about where discarded electronics are accumulating, how much waste is being generated, or when a particular locality will require a collection service. This can contribute to delays, inefficient collection routes, temporary accumulation and increased chances of improper disposal.

This report proposes E-Predict, a web-based predictive e-waste collection and smart pickup management system. The platform allows users, institutions and collection partners to report the type, approximate quantity and location of e-waste awaiting collection. The system aggregates this information and applies a demand-prediction and priority model to estimate collection requirements for different areas. A dashboard can then recommend which locations should be collected first and when a pickup should be scheduled.

The proposed system is designed as a decision-support platform rather than a replacement for authorized recycling operations. Its purpose is to make e-waste collection proactive and data-driven. The solution supports Sustainable Development Goal 12 on responsible consumption and production, SDG 11 on sustainable cities and waste management, and indirectly SDG 13 on climate action. Its environmental management relevance is examined using the life-cycle perspective of ISO 14001, while a simplified life-cycle assessment framework is used to identify the stages at which the proposed system can influence environmental impacts.

The report evaluates the proposed system from technical, economic, operational, social and environmental perspectives and discusses limitations such as data accuracy, prediction uncertainty, privacy, user participation and integration with authorized recyclers.

1. Introduction

Electronic devices have become an essential part of modern life. Smartphones, laptops, tablets, televisions, networking equipment, printers, household appliances and other products are replaced or discarded as technology changes or as devices become damaged or obsolete. Although these products provide significant social and economic benefits, their end-of-life management creates an important sustainability challenge.

E-waste is different from ordinary municipal waste because electronic products may contain both valuable recoverable materials and substances that require controlled handling. The Central Pollution Control Board (CPCB) notes that e-waste can contain valuable metals as well as hazardous substances and that its management in India is regulated under the E-Waste (Management) Rules, 2022, which took effect on 1 April 2023 [2].

The scale of the global problem is substantial. According to UNITAR and ITU, the world generated 62 million tonnes of e-waste in 2022, and documented environmentally sound collection and recycling accounted for only 22.3% of that quantity. The gap between generation and documented recycling is expected to widen without stronger management systems [1].

Existing digital solutions can support individual parts of the e-waste process. For example, a user may be given information about recycling, a collection location, a reward for handing over waste, or a way to reuse an old device. However, the proposed solution in this report focuses on a different operational gap: the lack of a predictive, demand-based view of where collection capacity is needed.

E-Predict therefore treats e-waste collection as a data and logistics problem. Instead of waiting until waste has accumulated and then responding, the platform collects early reports and historical data, estimates future demand and supports smarter scheduling. The system is intended to complement authorized collection and recycling infrastructure.

1.1 Objectives

- To develop a web-based system for collecting structured information about e-waste awaiting pickup.
- To estimate collection demand using historical and current user-reported data.
- To identify areas or institutions with high, medium or low collection priority.
- To support data-driven pickup scheduling and route planning.
- To provide dashboards that help collection partners understand e-waste generation patterns.
- To reduce the likelihood of prolonged accumulation and improper disposal by enabling earlier intervention.
- To generate useful aggregate data for institutions, local bodies and recycling partners without exposing unnecessary personal information.

1.2 Scope

The student prototype focuses on reporting, aggregation, prediction, prioritization and pickup scheduling. It does not itself perform physical collection, dismantling or recycling. The final handling of e-waste must remain with appropriate authorized entities. The prediction model may begin with a rule-based or simple statistical approach and can later be upgraded to machine-learning models when sufficient historical data become available.

2. Problem Statement

Improper disposal of electronic waste is a growing environmental problem caused not only by a lack of awareness but also by weaknesses in the collection process. When e-waste is stored for long periods, mixed with ordinary waste, abandoned or handed to inappropriate handlers, opportunities for safe recovery of materials are reduced and environmental risks can increase.

A key operational difficulty is the limited availability of timely, location-specific information about e-waste generation. Collection systems may operate according to fixed schedules or respond only after users request a service. Such approaches do not necessarily reflect changes in local demand. For example, an educational institution may generate a large amount of obsolete electronics near the end of an academic year, while a residential area may experience a different pattern. Without data, collection capacity cannot easily be matched to these changing patterns.

Therefore, the engineering problem addressed by this project is:

“How can information technology be used to predict e-waste collection demand and support timely, efficient and environmentally responsible pickup scheduling?”

2.1 Key Problems Identified

Problem	Effect	E-Predict Response
Collection is often reactive	Pickup may occur only after waste has accumulated	Early reporting and demand prediction
Limited visibility of quantity	Collection capacity may not	Quantity/weight estimates stored

	match actual demand	in the system
Changing spatial demand	Some areas may need urgent collection while others do not	Area-wise priority scoring
Fixed or inefficient schedules	Unnecessary trips or delayed pickups can occur	Recommended pickup timing
Limited historical data	Future demand is difficult to anticipate	Historical reports used for trend analysis
Poor coordination	Users and collection partners may lack a common information source	Shared dashboards and status updates

2.2 Uniqueness of the Proposed Solution

The proposed solution is deliberately different from other common e-waste website concepts. It does not primarily provide rewards, decide whether a device should be repaired or recycled, display a map of collection centres, create a digital ownership history for each device, or suggest second-life uses. Its central digital object is the collection demand itself: the system asks where e-waste is accumulating, how much is being reported, how demand is changing and when a pickup should be prioritized.

Alternative Approach	Primary Focus	E-Predict Difference
Reward platform	Motivate users through points or benefits	Predicts collection demand rather than rewarding disposal
Reuse/repair advisor	Suggest what to do with an old device	Focuses on collection capacity and timing
Collection-point map	Show where facilities are located	Predicts where collection is needed
Digital device passport	Track an individual device through its life	Tracks aggregated collection demand, not device ownership history
Second-life platform	Find alternative uses for old electronics	Optimizes collection response before disposal

3. Sustainability Analysis

3.1 Relevant Sustainable Development Goals

SDG 12 – Responsible Consumption and Production

SDG 12 is the strongest alignment because it promotes sustainable consumption and production and the sustainable management of resources [3]. E-Predict supports this goal by improving the organization of e-waste collection, increasing the likelihood that discarded electronics enter formal management channels and generating data that can support better resource recovery.

SDG 11 – Sustainable Cities and Communities

Target 11.6 calls for reducing the adverse environmental impact of cities, including attention to air quality and municipal waste management [4]. A demand-based e-waste collection system can support organized urban waste workflows by identifying locations where electronic waste requires attention and helping collection partners allocate resources.

SDG 13 – Climate Action

The relationship with SDG 13 is indirect. Better collection can increase the likelihood that materials are recovered rather than lost and can support more efficient waste logistics. Avoiding unnecessary replacement and improving material recovery can reduce some upstream environmental pressures, although this prototype does not claim a direct quantified carbon reduction.

3.2 ISO 14001 Relevance

ISO 14001 provides a framework for environmental management systems and emphasizes systematic management of environmental responsibilities and continual improvement. The current edition was updated in 2026 [5]. A key concept relevant to this project is the life-cycle perspective: environmental aspects should be considered across stages that an organization can control or influence, including end-of-life treatment and final disposal [6].

E-Predict can support an organization applying an environmental management system in the following way:

ISO 14001-oriented principle	Application in E-Predict
Identify environmental aspects	Identify e-waste generation and accumulation as an environmental aspect requiring management.
Operational planning and control	Use demand data to plan collection activities and reduce uncontrolled accumulation.
Performance monitoring	Track reports, collection response time, quantities collected and area-wise trends.
Continual improvement	Use historical data to refine prediction rules and collection schedules.
Life-cycle perspective	Focus on the end-of-life stage and its connection with resource recovery and disposal.
Compliance support	Design integration with authorized collection/recycling entities and applicable Indian e-waste requirements.

3.3 Life Cycle Assessment (LCA) Perspective

ISO 14040 defines the principles and framework for life cycle assessment, including goal and scope definition, life-cycle inventory, impact assessment and interpretation [7]. ISO 14044 provides requirements and guidelines for conducting LCA studies [8]. A complete LCA of an e-waste management system would require detailed primary data and is outside the scope of a student prototype. However, a simplified life-cycle perspective can be used to identify where E-Predict can influence environmental outcomes.

Life-cycle stage	Environmental concern	Potential contribution of E-Predict
Raw material extraction	Resource use and environmental impacts	Indirectly supports material recovery by improving formal collection.
Manufacturing	Energy and material consumption	Indirectly supports circularity by helping retain recoverable materials.
Distribution	Transport and logistics impacts	Collection demand data can support better scheduling and fewer avoidable trips.
Use	Product replacement and waste generation	Historical demand data can reveal recurring generation patterns.
End-of-life collection	Accumulation, leakage and improper disposal	Primary intervention point: predict demand and prioritize pickup.
Recycling/recovery	Safe processing and material recovery	System can confirm collection handover and record aggregate outcomes.

3.4 Environmental Impact Pathway

The proposed environmental pathway is: better information → earlier identification of collection demand → timely pickup → improved channelization to formal management → greater opportunity for safe recycling and resource recovery → reduced likelihood of improper disposal. The system itself does not recycle e-waste; its sustainability value comes from improving the information layer that supports the physical waste-management process.

4. Proposed Engineering/IT Solution

E-Predict is a responsive web application that connects e-waste generators with collection-management personnel through a shared data platform. Users report e-waste that is ready or likely to become ready for collection. The platform records category, approximate quantity, location and requested timeframe. The system then calculates a collection priority and uses historical data to estimate future demand.

4.1 System Components

1. User Reporting Module – records e-waste reports from households, students, offices, institutions or other participating users.
2. Data Management Module – stores structured reports, timestamps, locations, quantities and collection status.

3. Prediction Engine – analyzes historical and current data to estimate near-future collection demand.
4. Priority Engine – classifies locations or requests into priority levels using quantity, age of report, trend and capacity-related rules.
5. Smart Scheduling Module – recommends pickup dates and groups requests that can be served efficiently.
6. Admin/Collector Dashboard – displays demand hotspots, pending requests, predicted demand and recommended actions.
7. Analytics Module – provides trends such as monthly reports, category distribution, collected quantity and response time.
8. Notification Module – can notify users when a pickup is scheduled or when collection status changes.

4.2 Basic System Workflow



4.3 User Module

A user creates an account or uses an institution-managed interface. The user submits an e-waste report containing the broad device category, approximate number of items, estimated weight if known, locality or approved collection zone, and the desired collection timeframe. The user can also view the status of the report.

Input	Example
Category	Laptops
Quantity	6
Estimated weight	15 kg
Location/zone	College Campus – Block A
Expected availability	Within 3 days
Status	Awaiting pickup

4.4 Prediction Engine

The prediction engine is the main differentiating feature of the proposed system. In an initial prototype, it can use a transparent rule-based or moving-average model rather than a complex machine-learning model. For

example, the system can calculate the average quantity reported in a zone over recent periods and compare it with the current reporting rate.

A simple moving-average estimate can be expressed as:

$$\text{Predicted Demand} = (D_1 + D_2 + \dots + D_n) / n$$

where D represents the observed e-waste quantity for each previous period and n is the number of periods considered. A later version can incorporate additional features such as seasonality, institution calendars, population, historical response time and category-specific trends using machine-learning regression or time-series models.

The prototype should clearly label predictions as estimates. Prediction output should support decision-making rather than be presented as a guaranteed future quantity.

4.5 Priority Scoring

A priority score can combine quantity, age of request and predicted demand. One possible prototype model is:

$$\text{Priority Score} = w_Q Q + w_A A + w_T T$$

where Q represents normalized reported quantity, A represents the age of pending reports, T represents the local demand trend, and w_Q , w_A and w_T are adjustable weights. The exact weights would be calibrated during testing. This makes the system explainable and easier to validate than an opaque model.

Priority	Typical condition	Suggested action
High	Large quantity, old pending request or rapidly increasing demand	Schedule earliest feasible pickup
Medium	Moderate quantity or steady demand	Schedule in normal collection cycle
Low	Small quantity and low predicted demand	Combine with nearby/future collection where practical

5. System Architecture and Implementation

5.1 Three-Tier Architecture

Layer	Suggested Technology	Purpose
Presentation	HTML, CSS, JavaScript / React	Responsive forms, dashboards, charts and status views
Application	Node.js + Express or Python Flask	Business logic, prediction, priority calculation and APIs
Data	MySQL	Users, reports, collection requests, predictions and status records
Analytics	JavaScript charts / Recharts / Chart.js	Demand trends and dashboard visualization
Authentication	Session/JWT + password hashing	Secure account and role management

Deployment	Cloud hosting / college server	Web access for prototype or pilot
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5.2 Main Database Entities

Table	Important Fields	Purpose
users	id, name, email, role	User and administrator identity
ewaste_reports	id, user_id, category, quantity, weight, zone, reported_at, status	Individual e-waste reports
zones	id, name, location, capacity	Collection areas or institutional zones
demand_history	id, zone_id, period, quantity	Historical data used for prediction
predictions	id, zone_id, period, predicted_quantity, generated_at	Stored demand estimates
pickup_requests	id, zone_id, priority, scheduled_date, status	Collection scheduling
collections	id, pickup_id, collected_weight, collected_at	Actual collection outcome
notifications	id, user_id, message, sent_at, status	Pickup/status communication

5.3 Data Flow

9. The user submits an e-waste report.
10. The application validates the category, quantity, zone and date fields.
11. The report is stored in the database with a timestamp.
12. The system aggregates reports by zone and time period.
13. The prediction engine calculates expected near-future demand.
14. The priority engine assigns collection urgency.
15. The dashboard displays recommended pickup actions.
16. A collector/admin confirms or modifies the proposed schedule.
17. After pickup, the actual collected quantity is entered.
18. The actual outcome becomes historical data for future prediction.

5.4 Security and Privacy

- Collect only information necessary for collection planning.
- Avoid exposing precise personal addresses publicly; use approved zones where possible.
- Use password hashing rather than storing plain-text passwords.
- Use role-based access for user, collector and administrator functions.
- Validate inputs on both client and server sides.
- Protect the database and API against unauthorized access.
- Record administrative changes for accountability.
- Use aggregated data in public analytics to reduce privacy risks.

5.5 Responsive Website Pages

Page	Main Content
Home	Problem overview, project purpose and call to action
Report E-Waste	Form for category, quantity, weight, zone and collection timeframe
My Reports	Submitted reports and pickup status
Demand Dashboard	Current demand, predicted demand and priority areas
Pickup Management	Recommended and scheduled collections
Analytics	Charts showing trends, categories and collected quantities
Awareness	Safe handling and responsible e-waste information
Admin Login	Restricted management interface

6. Feasibility Analysis

6.1 Technical Feasibility

The proposed platform is technically feasible as a student web application. A minimum viable prototype can be implemented using common web technologies and a relational database. The prediction engine does not require advanced artificial intelligence in its first version. A moving average, weighted score or simple regression model can demonstrate the concept. Once sufficient real-world data are available, more advanced time-series or machine-learning methods can be evaluated.

The responsive interface can be implemented using standard HTML/CSS/JavaScript or a framework such as React. A backend such as Node.js with Express or Python with Flask can expose REST APIs. MySQL is suitable for storing structured reports and historical data.

6.2 Economic Feasibility

The prototype can be developed using free or open-source development tools. The main costs in a real deployment would come from hosting, database infrastructure, security, maintenance, notification services, mapping or routing services if required, and integration with collection partners. Because the system is software-based, it does not require specialized physical hardware for the initial implementation.

Cost Area	Prototype Requirement	Full Deployment Consideration
Software	Mostly free/open-source	Licensing only where required
Hosting	Low-cost/free student hosting	Scalable cloud infrastructure
Database	Local/MySQL server	Managed database and backups
Notifications	Optional/demo	SMS/email/push service costs
AI/analytics	Simple model	Compute and model-development costs if scaled
Maintenance	Student/team effort	Dedicated technical support

6.3 Operational Feasibility

The workflow is simple enough for non-technical users: report e-waste, wait for scheduling information, hand over the material and view the updated status. For collection personnel, the dashboard reduces the need to manually interpret scattered requests. The system can initially operate within a college campus, residential community or small municipality before being expanded.

6.4 Social Feasibility

The system can encourage responsible disposal by making collection more convenient and predictable. Users receive visibility into whether a request has been received and scheduled. Institutions can use the analytics to organize periodic drives or targeted collection campaigns. The platform also creates an opportunity to communicate safe handling practices for batteries and other potentially hazardous components.

6.5 Environmental Feasibility

The solution does not require a new physical treatment process. Its environmental benefit depends on whether better information leads to earlier collection and better channelization to appropriate handlers. Therefore, the system should be integrated with authorized collection and recycling partners rather than presenting itself as a recycler.

6.6 Risk Analysis

Risk	Potential Effect	Mitigation
Inaccurate user quantity estimates	Prediction error	Use ranges/validation and treat estimates as approximate
Low user participation	Insufficient data	Institutional campaigns and simple reporting interface
Insufficient historical data	Weak prediction	Start with rule-based scoring and improve over time
False or duplicate reports	Inflated demand	Authentication, validation and duplicate checks
Privacy concerns	Loss of user trust	Zone-level reporting and minimal data collection
Collector does not follow recommendation	Operational mismatch	Allow human override and record actual outcomes
Unauthorized recycling partner	Environmental/legal risk	Integrate only with verified/authorized entities

7. Expected Environmental, Social and Economic Benefits

7.1 Environmental Benefits

- Reduced probability of e-waste remaining uncollected for long periods.
- Improved channelization of reported e-waste toward formal collection and recycling systems.
- Better opportunity for recovery of valuable materials from discarded electronics.
- Potential reduction in unnecessary collection trips through demand-based scheduling and grouping.
- Improved visibility of recurring e-waste generation patterns, supporting targeted prevention and collection campaigns.

7.2 Social Benefits

- Provides users with a simple way to report e-waste requiring collection.
- Improves transparency about collection status.
- Supports institutions and communities in organizing e-waste drives.
- Creates awareness about responsible handling and disposal.
- Can support safer waste-management practices by reducing dependence on uncontrolled disposal.

7.3 Economic Benefits

- Better scheduling can improve utilization of collection resources.
- Recovered materials may retain economic value when processed through appropriate recycling systems.
- Historical data can help organizations estimate collection capacity requirements.
- A scalable software platform can support multiple institutions or local collection zones without requiring a completely new system for each location.

8. Justification of the Proposed Solution

The proposed solution is justified because the e-waste problem has both a physical and an information dimension. Physical recycling infrastructure is necessary, but its effectiveness can be limited when collection demand is poorly understood. A collection partner that knows the approximate quantity and location of pending e-waste can plan resources more effectively than one that receives scattered requests without historical context.

E-Predict addresses this information gap by converting individual reports into an aggregated demand picture. The platform does not attempt to replace recycling facilities or existing regulatory systems. Instead, it acts as an information and decision-support layer between waste generators and collection operations.

The solution is also appropriate for an engineering/IT project because it combines database design, responsive web development, data analytics, prediction, prioritization and system workflow design. The concept can begin with a small college-based dataset and be expanded gradually. This makes it both demonstrable as a student prototype and potentially scalable to larger communities.

A further advantage is explainability. The first version can use transparent prediction and priority rules rather than a black-box model. This allows the team to show exactly why an area was marked high priority. Machine learning can be introduced later when enough reliable historical data are available.

8.1 Example Scenario

Consider a college campus during the end of an academic year. Several laboratories, departments and students may simultaneously discard old chargers, keyboards, monitors, laptops and other electronics. Individual users submit reports through E-Predict. The system aggregates the reports and detects that one campus zone has a rapidly increasing reported quantity.

The dashboard may show that the zone has a high collection priority and recommend a pickup within the next available collection window. The collector can then group nearby requests into the same trip. After collection, the actual quantity is recorded. This information becomes part of the historical dataset and improves future predictions for similar periods.

The scenario demonstrates the central idea: instead of waiting for waste to accumulate and then reacting, the system uses early information to anticipate collection demand.

9. Conclusion

Improper disposal of electronic waste is a significant sustainability challenge because electronic products contain both recoverable resources and materials that require responsible handling. Global e-waste generation is increasing faster than documented formal recycling, creating a need for improved management systems [1]. In India, e-waste management is governed by the E-Waste (Management) Rules, 2022, which emphasize environmentally sound management and extended producer responsibility [2].

This report proposed E-Predict, an AI-assisted predictive e-waste collection and smart pickup management system. Unlike solutions that focus primarily on rewards, disposal advice, facility maps, individual device histories or second-life applications, E-Predict focuses on the operational question of when and where collection demand will arise.

The system collects structured e-waste reports, aggregates them by area and time, estimates future demand, calculates collection priority and recommends pickup schedules. The architecture can be implemented using a responsive web frontend, backend APIs, a MySQL database and a transparent prediction model. The system can later be enhanced with machine-learning methods as the volume and quality of historical data improve.

From a sustainability perspective, the strongest alignment is with SDG 12 because the platform supports responsible waste management and resource recovery. It also contributes to SDG 11 through better urban waste-management workflows and has an indirect relationship with SDG 13. The system is compatible with the life-cycle perspective emphasized by ISO 14001 because it targets the end-of-life stage where collection and disposal decisions influence environmental outcomes [5][6].

The proposed system cannot by itself guarantee responsible recycling. Its success depends on user participation, data quality, reliable collection partners and compliance with applicable regulations. Nevertheless, by improving the information available before collection takes place, E-Predict provides a practical IT-based approach to making e-waste management more proactive, measurable and efficient.

9.1 Future Scope

- Replace the initial prediction model with time-series or machine-learning models after sufficient historical data are collected.
- Integrate verified recycler and collection-partner systems.
- Add route optimization to combine geographically compatible pickup requests.
- Develop a mobile application for faster reporting.
- Use barcode/QR scanning to simplify category entry without creating a device-lifecycle passport.
- Add institutional analytics for schools, colleges, offices and municipalities.
- Introduce anomaly detection to identify unusual spikes or duplicate reports.
- Provide multilingual interfaces for broader accessibility.
- Integrate official reporting and compliance workflows where technically and legally appropriate.

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